Sample Paper on Diffusion Distillation

Abstract. We study distillation of diffusion models into few-step generators. Our method reduces sampling cost by a factor of eight while keeping image quality.

1 Introduction

Diffusion models produce high quality samples but require many network evaluations. Distillation compresses the sampling trajectory into a small number of steps. We propose a progressive scheme that halves the number of steps at every stage.

2 Method

A student network is trained to match two teacher steps with one student step. The procedure is repeated until the student needs only four steps in total.

3 Experiments

We evaluate on ImageNet and report FID against the undistilled teacher baseline. The distilled model achieves an eight times speedup with a small FID increase. Ablations show that the loss weighting matters more than the schedule choice.

4 Conclusion

Progressive distillation offers a simple recipe for fast sampling from diffusion models.

References

[1] Ho et al. Denoising diffusion probabilistic models.